GLOBAL FUEL PRICES AND TAXATION TRENDS

A Comprehensive Global Performance Intelligence Report

SUBMISSION DETAILS	
Student Name	ABIR MENDHALKAR
Individual ID	2020–2026	84 countries	7 regions
Project Scope	27,468 records	10 variables
Dataset Size	1 November 2026
Prepared	

*Self-assigned individual working ID because no team ID was supplied. It is not an institution-issued identifier.

Consolidated, completed project-template pack

This document consolidates the supplied ideation, requirements, design, planning, data-preparation, visualization, dashboard, performance-testing, demonstration and final-report templates. It uses the project description already provided. Actual dataset files, Snowflake execution logs, Tableau screenshots, repository links and demonstration evidence were not included in the ZIP; those items are explicitly identified rather than fabricated.

Submission notes and document index

•All28files in the ZIPwerereviewed. Duplicate/alternateversionsof the Problem–Solution Fit, Business Question & Visualisation, and Dashboard Design templates are consolidated into the matching sections below.

• The ZIP already contains templates for the major listed deliverables. No additional template could be reliably identified as missing from the supplied package, so no unrelated third-party template has been substituted.

• The supplied dataset description is treated as the project brief, not as proof that the dataset was loaded or tested.

• Replace any “to be captured” evidence items with genuine screenshots, measured timings, SQL outputs, Tableau Public links and repository details before final academic submission.

Supplied template file(s)	Completed section in this PDF
Brainstorm & Idea Prioritization Template.docx	2.3 Brainstorming and prioritization
Define Problem Statements Template.docx	2.1 Customer problem statements
Empathy Map Canvas.docx	2.2 Empathy map
Customer Journey Map.pdf	3.1 Customer journey
Problem–Solution Fit Template v1.docx/.pdf and Sample.pdf	4.1 Problem–solution fit
Project Proposal template.docx	4.2 Proposed solution / proposal
Solution Requirements.docx	3.2 Functional and non-functional requirements
Data Flow Diagrams and User Stories.docx	3.3 DFD and user stories
Technology Stack - Template.docx	3.4 Technology stack
Solution Architecture.docx	4.3 Solution architecture
Project Planning Template.docx	5.1 Project planning
Data Preparation Template.docx	6.1 Data preparation
Business Question and Visualisation Report.docx (both copies)	7.1 Eight business questions and visualisations
Dashboard Design Template.docx (both copies)	7.2 Three dashboard designs
Story Design Template.docx	7.3 Tableau story design
Snowflake .docx and Tableau .docx	8. Performance testing plans
Code & Solution.docx	9. Code and solution summary
Final Report Template.pdf	10. Consolidated final report
Communication.pdf, Demonstration of Proposed Features.pdf, Project Demo Planning.pdf, Scalability & Future Plan.pdf, Team Involvement in Demonstration.pdf	11. Communication, demo, scalability and future plan

1. Introduction

Field	Value
Student / Individual	ABIR MENDHALKAR
Individual ID	IND-ABIR-055 (self-assigned working ID; replace if institution issues an official ID)
Project	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Project period	2020–2026
Dataset scope	27,468 records • 84 countries • 7 global regions • 10 variables (as supplied in project brief)
Prepared on	1 November 2026

1.1 Project overview

The project analyses global petrol, diesel and LPG prices alongside income levels, subsidy structures and fuel-tax policies across 84 countries and 7 global regions from 2020 to 2026. The supplied brief describes 27,468 complete records and 10 variables. The intended output is an interactive business-intelligence solution that helps users compare fuel-price levels, examine changes over time, and explore how taxes, subsidies and income context relate to observed prices.

1.2 Purpose

• Provide a consistent view of fuel-price differences across countries and regions.

• Track yearly and fuel-type trends from 2020 through 2026.

• Compare petrol, diesel and LPG using common filters and measures.

• Support policy, research and planning discussions with transparent visual evidence.

• Make the analysis accessible through Tableau dashboards and a guided data story.

1.3 Intended audience

• Energy-policy analysts and public-sector decision makers.

• Researchers, educators and students studying energy economics.

• Fuel-market and sustainability analysts.

• Journalists and informed citizens seeking comparable country-level context.

Scope note: This is a descriptive BI project. Correlation or co-movement between taxes, subsidies, income and prices must not be presented as proof of causation.

2. Ideation Phase

Field	Value
Student / Individual	ABIR MENDHALKAR
Individual ID	IND-ABIR-055 (self-assigned working ID; replace if institution issues an official ID)
Project	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Project period	2020–2026
Dataset scope	27,468 records • 84 countries • 7 global regions • 10 variables (as supplied in project brief)
Prepared on	1 November 2026

2.1 Customer problem statements

ID	I am (customer)	I'm trying to	But	Because	Which makes me feel
PS-1	An energy-policy analyst	Compare fuel prices and tax/subsidy policies across countries and years	Data is spread across multiple measures and countries	Definitions, currencies, units and policy structures differ	Uncertain about making fair comparisons
PS-2	A researcher or student	Understand how fuel prices vary by fuel type, region and income context	Static tables make patterns difficult to spot	Large multi-year datasets are hard to explore manually	Overwhelmed and time-constrained
PS-3	A journalist / public information user	Find clear, defensible fuel-price trends	Headline figures can hide regional and temporal variation	There is no single interactive view tailored to the questions	At risk of oversimplifying the story

2.2 Empathy map canvas

Perspective	User insight
SAYS	"I need comparable figures by country, fuel type and year." "Show me the source, units and filters used."
THINKS	Are the currencies and units consistent? Are taxes and subsidies measured comparably? Is a trend caused by policy or by wider market conditions?
DOES	Downloads spreadsheets, searches official energy and government sources, builds pivot tables, compares charts and prepares briefing slides.
FEELS	Frustrated by inconsistent definitions; cautious about drawing conclusions; more confident when methods, filters and caveats are visible.
PAINS	Scattered sources, missing values, changing currencies/units, policy terminology differences, time-consuming comparisons.
GAINS	A trusted, filterable dashboard; clear trend charts; country rankings; transparent definitions; exportable findings.

These are research-informed user hypotheses for the design exercise, not quotations from interviews. Validate them through interviews or usability testing before treating them as measured customer evidence.

2.3 Brainstorming and idea prioritization

No.	Idea / suggestion	Category	Feasibility	Importance	Priority	Selected
1	Interactive country and region comparison dashboard	Visualization	High	High	1	Yes
2	Fuel-type selector for Petrol / Diesel / LPG	Interaction	High	High	2	Yes
3	Year trend analysis from 2020–2026	Time series	High	High	3	Yes
4	Tax and subsidy comparison views	Policy analysis	Medium	High	4	Yes
5	Country ranking for highest/lowest fuel prices	Benchmarking	High	Medium	5	Yes

No.	Idea / suggestion	Category	Feasibility	Importance	Priority	Selected
6	Forecasting future prices	Advanced analytics	Low	Medium	6	Future scope
7	Automated refresh from official APIs	Data engineering	Medium	High	7	Future scope

Selected concept: a filter-driven global fuel-price intelligence dashboard with trend, ranking, regional and policy-context views. It offers high feasibility using the described structured dataset and directly addresses the comparison problem.

3. Requirement Analysis

Field	Value
Student / Individual	ABIR MENDHALKAR
Individual ID	IND-ABIR-055(self-assigned working ID; replace if institution issues an official ID)
Project	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Project period	2020–2026
Dataset scope	27,468 records • 84 countries • 7 global regions • 10 variables (as supplied in project brief)
Prepared on	1 November 2026

3.1 Customer journey map

Stage	User action	Pain point	Opportunity / system response	Success signal
Discover	Searches fuel prices or policy comparisons	Sources and definitions vary	Landing page explains coverage, measures and sources	User understands scope
Access	Opens dashboard and selects country/region/year	Too many filters can confuse	Provide clear defaults and concise filter labels	User reaches a relevant view quickly
Explore	Changes fuel type and compares countries	Manual comparison is slow	Fuel Type parameter and country/region filters	Charts update consistently
Interpret	Reviews trends, rankings and policy context	Risk of confusing correlation with causation	Show units, year, definitions and caveats	User can explain observed pattern accurately
Share / decide	Exports or presents findings	Static screenshots lose filter context	Provide captions, source notes and export guidance	Findings are reproducible

3.2 Functional requirements

FR	Epic	Sub-requirement	Priority
FR-1	Data access & integration	Load the 2020–2026 fuel dataset containing country, region, fuel prices, income, subsidy and tax fields; preserve source metadata and units.	High
FR-2	Data processing	Profile nulls, duplicates, data types, inconsistent country/region labels, currency/unit formats and invalid ranges; document all treatment.	High
FR-3	Business analysis	Calculate average fuel price by fuel type, country, region and year; compare price changes and rank countries.	High
FR-4	Insights & outputs	Provide at least 8 analytical worksheets, 3 dashboards and a Tableau story with clear captions.	High
FR-5	User interaction	Allow selection of Petrol, Diesel or LPG; filter by year, country and region; support comparisons and tooltips.	High
FR-6	Update & reporting	Allow refresh/replacement of source data and export of dashboard views; display last-refresh/source note when available.	Medium

3.2.1 Non-functional requirements

ID	Requirement	Description
NFR-1	Usability	Consistent labels, readable charts, accessible contrast, meaningful defaults and clear filter controls.
NFR-2	Security	Use authorized data sources; do not expose credentials; restrict access to private data and use least privilege.
NFR-3	Reliability	Document repeatable data preparation and validate totals, ranges and filter behaviour after refresh.
NFR-4	Performance	Keep worksheets focused, reduce unnecessary marks and test dashboard responsiveness on the target device.
NFR-5	Availability	Publish through an approved Tableau environment; availability depends on hosting and account settings.

ID	Requirement	Description
NFR-6	Scalability	Use efficient extracts or a governed warehouse and modular calculations to support more countries and years.

3.3 Data flow diagram (textual Level-1 representation)

Source / entity	Process	Data store / output	Flow
CSV / Excel source dataset	1.0 Ingest data	RAW_FUEL_PRICES	Source records + metadata
RAW_FUEL_PRICES	2.0 Profile & clean	STG_FUEL_PRICES	Type checks, null/duplicate checks, standardization
STG_FUEL_PRICES	3.0 Transform & calculate	CURATED_FUEL_PRICES / analytical views	Derived year, fuel type, average price and comparison measures Filtered measures and aggregated results
CURATED data / views	4.0 Visualize	Tableau worksheets, dashboards and story	Selections, comparisons and findings
Analyst / student / policy user	5.0 Explore & report	Dashboard exports / report	

External entities: dataset provider and dashboard user. Processes: ingest, clean, transform, visualize and report. Data stores: raw, staging and curated tables/views.

3.3.1 User stories

User type	Story ID	User story	Acceptance criteria	Priority
Analyst	US-01	As an analyst, I can choose Petrol, Diesel or LPG.	All relevant views update to the selected fuel type.	High
Analyst	US-02	As an analyst, I can filter by year and region.	Only records matching selected filters are reflected. Country values and units are clearly displayed.	High
Researcher	US-03	As a researcher, I can compare countries.	The chart displays the 2020–2026 period where data exists.	High
Researcher	US-04	As a researcher, I can view yearly trends.	Related measures appear with definitions and caveats.	High
Policy user	US-05	As a policy user, I can compare price, tax and subsidy context.	Each dashboard includes scope, units and source notes.	Medium
Viewer	US-06	As a viewer, I can read explanatory captions and source notes.	Export is available where supported by the publishing environment.	High
Viewer	US-07	As a viewer, I can export a view for reporting.		Medium

3.4 Technology stack

Component	Selected technology	Purpose / justification
Programming / query language	SQL; optional Python for preprocessing; calculated fields in Tableau	SQL supports structured transformations; Python can automate profiling; Tableau calculations support interactive analysis.
Data store	CSV / Excel source; Snowflake as optional warehouse layer	The provided brief describes structured data; use Snowflake if the course environment requires warehouse ingestion.
Data ingestion	Tableau data connection; Snowflake stage + COPY INTO if Snowflake is used	Load the source reproducibly and retain a raw copy.
Transformation	SQL / Tableau Prep / Tableau calculated fields	Clean and standardize data, derive measures and build reusable analytical fields.
Visualization	Tableau Desktop / Tableau Public	Create interactive worksheets, dashboards and story points.
Interface	Tableau dashboards; optional Flask or Streamlit wrapper only if implemented	Provide filter and parameter controls without implying a separate app already exists.

4. Project Design

Field	Value
Student / Individual	ABIR MENDHALKAR
Individual ID	IND-ABIR-055(self-assigned working ID; replace if institution issues an official ID)
Project	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Project period	2020–2026
Dataset scope	27,468 records • 84 countries • 7 global regions • 10 variables (as supplied in project brief)
Prepared on	1 November 2026

4.1 Problem–solution fit canvas

Canvas element	Project answer
Customer segments	Energy-policy analysts, researchers, students, journalists and public-information users.
Jobs-to-be-done / problems	Compare prices across countries, fuels, regions and years; identify high/low price markets; understand tax/subsidy context.
Triggers	Policy announcements, fuel-price changes, research assignments, budget planning, news events and requests for country benchmarking.
Emotions before / after	Before: uncertainty, effort and frustration. After: clearer comparisons and more confidence, provided source and limitations are visible.
Available solutions	Official government portals, international energy datasets, spreadsheets, static reports and individual country dashboards.
Customer constraints	Different units/currencies, data availability, varying definitions, limited time, limited BI skills and internet/device constraints.
Behaviour	Searches official sources, downloads spreadsheets, compares figures manually, creates charts and shares reports.
Online channels	Government and energy-statistics websites, public datasets, research portals and interactive dashboards.
Offline channels	Policy briefings, classrooms, research meetings, presentations and printed reports.
Root causes	Fragmented information, non-uniform reporting practices, complex policy measures and lack of a single comparison interface.
Proposed solution	A filterable Tableau BI solution with country/region/year comparisons, fuel selector, ranking charts, trend views and policy-context analysis.

Fit assessment: the proposed dashboard aligns with the user's recurring comparison and reporting tasks. Actual product–market fit is not established; it requires user testing and feedback.

4.2 Project proposal

Objective

Design and implement a descriptive business-intelligence report that makes global fuel-price and taxation-related data easier to compare and interpret across countries, regions, fuel types and years.

Scope

• In scope: dataset profiling, cleaning, derived analytical measures, eight visualizations, three dashboards, parameter/filter interactions and a Tableau story.

• Out of scope for the current version: causal econometric analysis, guaranteed future price forecasts, live market trading data and automated policy recommendations.

Problem description and impact

Users face difficulty comparing fuel prices because information is distributed across countries, years, fuel types and policy contexts. A consolidated view can reduce manual effort and improve the transparency of exploratory analysis.

Approach and key features

• Prepare a raw–staging–curated data workflow.

• Create a Fuel Type parameter with Petrol, Diesel and LPG options.

• Use an Average Fuel Price measure consistently across applicable views.

• Provide country rankings, regional comparisons, yearly trends and tax/subsidy context.

• Add clear tooltips, units, captions and source notes.

Resource requirements

Resource	Specification / allocation
Hardware	Standard laptop/desktop capable of running Tableau; no dedicated GPU required for the described BI workload.
Memory / storage	8 GB RAM recommended as a practical baseline; storage sized to source file, extracts and project files. Actual device specification not supplied.
Software	Tableau Desktop or Tableau Public; spreadsheet application; optional Snowflake account; optional Python environment.
Libraries / frameworks	Optional pandas and numpy for data profiling; not required if all preparation is done in SQL/Tableau.
Development environment	Tableau workbook (.twb/.twbx), SQL scripts where used, and a versioned project folder.
Data	Project brief states 27,468 records, 84 countries, 7 regions, 10 variables; source URL/file name not supplied.

4.3 Solution architecture

Layer	Component	Responsibility
Source	Fuel price and policy dataset	Provides country-year observations and fuel/tax/subsidy context.
Ingestion	CSV / Excel import or Snowflake stage	Loads the source without overwriting the original.
Raw	RAW_FUEL_PRICES	Preserves source columns and original values for traceability.
Staging	STG_FUEL_PRICES	Casts data types, standardizes labels, checks nulls/duplicates and flags invalid records.
Curated	CURATED_FUEL_PRICES	Stores validated fields and documented derived measures for analysis.
Analytics	Tableau data source + calculated fields	Computes averages, rankings, changes and aggregations based on selected dimensions.
Presentation	8 worksheets, 3 dashboards, 1 story	Enables exploration, comparison and communication of findings.
User	Analyst / researcher / student	Selects filters, interprets charts and exports findings.

Architecture note: The table is a logical architecture. The actual Snowflake database, warehouse, table names and deployment configuration must be updated to match the implemented environment.

5.1 Product backlog

Epic	Story ID	User story / task	Points	Priority	Sprint
Data ingestion	US-08	Load the source dataset and preserve a raw copy.	3	High	1
Data profiling	US-09	Profile nulls, duplicates, types, ranges and categories.	3	High	1
Data cleaning	US-10	Standardize country, region, year, fuel and policy fields.	5	High	1
Derived measures	US-11	Create Average Fuel Price and required comparison measures.	3	High	2
Trend analysis	US-12	Build yearly trend worksheets for selected fuel type.	3	High	2
Country ranking	US-13	Show top three countries by petrol price and comparable rankings.	2	High	2
Regional analysis	US-14	Compare regional averages and country variation.	3	Medium	2
Policy context	US-15	Visualize tax/subsidy and income context where fields permit.	5	Medium	3
Dashboard design	US-16	Build and format three dashboards with shared controls.	5	High	3
Story and QA	US-17	Create story points and validate filters, labels and colours.	3	High	4
Documentation	US-18	Prepare report, demo plan and limitations.	2	Medium	4

5.1.1 Proposed sprint schedule

Sprint	Focus	Duration	Planned story points	Exit criteria
Sprint 1	Ingestion, profiling and cleaning	1 week	11	Source loaded; quality checks documented; standardized staging data.
Sprint 2	Measures and core visualizations	1 week	11	Fuel selector, average price, trends and rankings implemented.
Sprint 3	Regional/policy analysis and dashboards	1 week	13	Three dashboard layouts assembled with filters and captions.
Sprint 4	Story, validation and documentation	1 week	5	Story sequence, QA checklist and final report prepared.

Planning estimate: 40 story points across four one-week sprints. This is a proposed plan, not a record of completed work. Individual work is assigned to SIDDHI SAMDANI; adjust points and dates to match actual progress.

Velocity and burndown method

Planned velocity = completed story points per sprint. Average daily velocity = completed points ÷ working days in that sprint. For a 5-working-day sprint with 10 completed points, the illustrative average is 2 points/day. Actual velocity must be calculated from the team's completed-work log.

Burndown: record remaining points at the start of each working day. Plot ideal remaining work against actual remaining work. No actual daily progress log was supplied, so no actual burndown values are asserted.

6. Data Preparation

Field	Value
Student / Individual	ABIR MENDHALKAR
Individual ID	IND-ABIR-055(self-assigned working ID; replace if institution issues an official ID)
Project	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Project period	2020–2026
Dataset scope	27,468 records • 84 countries • 7 global regions • 10 variables (as supplied in project brief)
Prepared on	1 November 2026

6.1 Data preparation workflow

Activity	Procedure	Validation / evidence to capture
1. Identify nulls	Profile null counts and percentages for each field. Decide whether to retain, impute, or exclude records based on field meaning; do not replace missing policy values with zero without justification.	SQL output showing null counts before/after treatment; documented decisions.
2. Handle duplicates	Define the natural key (for example country + year + fuel type, if appropriate to the source). Identify repeated keys and inspect whether they are true duplicates or distinct observations.	Duplicate-key query and row counts before/after deduplication.
3. Clean and standardize	Trim whitespace; standardize country/region/fuel labels; parse year and numeric fields; validate units/currency; flag impossible or out-of-range values.	Distinct-value checks, data types and exception report.
4. Transform and validate	Create a normalized fuel-type field and calculated measures; retain source values; validate record counts, ranges and aggregate totals.	Curated table schema, row counts, sample records and reconciliation.

6.1.1 Example Snowflake SQL patterns

The exactsourcecolumnnames werenotincludedintheZIP.The following are safe patterns to adapt after confirming the real schema; they are not claimed as executed queries.

```
-- 1) Inspect source columns and sample records
DESC TABLE RAW_FUEL_PRICES;
SELECT * FROM RAW_FUEL_PRICES LIMIT 20;

-- 2) Profile nulls (replace names with actual columns)
SELECT
  COUNT(*) AS total_rows,
  COUNT_IF(COUNTRY    IS    NULL)    AS    null_country,
  COUNT_IF(YEAR     IS     NULL)     AS     null_year,
  COUNT_IF(PETROL_PRICE  IS  NULL)  AS  null_petrol_price,
  COUNT_IF(DIESEL_PRICE  IS  NULL)  AS  null_diesel_price,
  COUNT_IF(LPG_PRICE IS NULL) AS null_lpg_price
FROM RAW_FUEL_PRICES;

-- 3) Find repeated candidate keys; validate key with source documentation
SELECT COUNTRY, YEAR, FUEL_TYPE, COUNT(*) AS row_count
FROM STG_FUEL_PRICES
GROUP BY COUNTRY, YEAR, FUEL_TYPE
HAVING COUNT(*) > 1;

-- 4) Check ranges and invalid years
SELECT *
FROM STG_FUEL_PRICES
WHERE YEAR NOT BETWEEN 2020 AND 2026
   OR PETROL_PRICE < 0
   OR DIESEL_PRICE < 0
   OR LPG_PRICE < 0;

-- 5) Create a curated view (adapt fields and table names)
CREATE OR REPLACE VIEW CURATED_FUEL_PRICES AS
SELECT
  TRIM(COUNTRY)  AS  COUNTRY,
  TRIM(REGION)   AS   REGION,
  YEAR,
  PETROL_PRICE,
  DIESEL_PRICE,
  LPG_PRICE,
  INCOME_LEVEL,
  SUBSIDY,
```

```
  TAX_POLICY
FROM STG_FUEL_PRICES
WHERE COUNTRY IS NOT NULL AND YEAR BETWEEN 2020 AND 2026;
```

6.1.2 Data-quality status

Check	Expected / supplied scope	Actual measured result
Rows	27,468 per project brief	Not independently verified; source file not included.
Countries	84 per project brief	Not independently verified.
Regions	7 per project brief	Not independently verified.
Years	2020–2026 per project brief	Not independently verified.
Variables	10 per project brief	Exact column names not supplied.
Nulls / duplicates / invalid values	Must be profiled before publication	Pending execution in source environment.

7. Business Questions and Visualizations

Field	Value
Student / Individual	ABIR MENDHALKAR
Individual ID	IND-ABIR-055 (self-assigned working ID; replace if institution issues an official ID)
Project	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Project period	2020–2026
Dataset scope	27,468 records • 84 countries • 7 global regions • 10 variables (as supplied in project brief)
Prepared on	1 November 2026

7.1 Eight business questions and proposed visualizations

#	Business question	Visualization	Fields / measure	Interpretation guide
1	How have petrol, diesel and LPG prices changed from 2020 to 2026?	Multi-line trend chart	Year; Average Fuel Price; Fuel Type	Compare direction and magnitude over time; use a fuel selector to avoid clutter.
2	Which three countries have the highest petrol prices?	Sorted horizontal bar chart (Top 3)	Country; Petrol Price	Show top three after sorting descending; display unit and selected year.
3	Which countries have the lowest petrol prices?	Sorted horizontal bar chart (Bottom 3)	Country; Petrol Price	Use ascending sort; ensure missing values are excluded transparently.
4	How do average fuel prices differ by global region?	Regional bar chart	Region; Average Fuel Price	Compare regional averages; show record/country coverage in tooltip if possible.
5	How do fuel prices vary across countries within each region?	Filled map or highlight table	Country; Region; Average Fuel Price	Reveal geographic variation; map availability depends on recognized country names.
6	How do tax-policy measures relate to fuel prices?	Scatter plot	Tax measure; Fuel Price; Country	Explore association only; do not infer causality from correlation.
7	How do subsidy structures compare across countries and fuel types?	Stacked bar / heatmap	Country; Subsidy category/value; Fuel Type	Use only if subsidy field is consistently coded and comparable.
8	How do income levels relate to fuel-price patterns?	Box plot or grouped bar	Income Level; Average Fuel Price	Compare distributions by income group; note that group definitions may vary.

The minimum of eight business questions is addressed as a design specification. Screenshots and numerical findings are not included because the Tableau workbook and data output were not supplied.

7.2 Dashboard designs (three dashboards)

Dashboard 1 — Global Fuel Price Overview

• KPI tiles: average petrol, diesel and LPG price (respecting selected filters).

• Yearly fuel-price trend line chart.

• Fuel Type parameter control: Petrol / Diesel / LPG.

• Filters: Year, Region and Country.

• Caption: “Values reflect the selected period, fuel type and available records.”

Dashboard 2 — Country and Regional Benchmarking

• Top 3 countries by petrol price and bottom 3 countries by petrol price.

• Regional average price comparison.

• Country map or highlight table.

• Filters: Year, Fuel Type and Region.

• Tooltips include country, region, selected fuel price and unit.

Dashboard 3 — Taxation, Subsidies and Affordability Context

• Scatter plot of tax measure versus selected fuel price, where compatible fields exist.

• Subsidy comparison by country/region.

• Income-level comparison using box plot or grouped bar.

• Filters: Year, Region and Fuel Type.

• Prominent note: descriptive association, not causal evidence.

Dashboard design principles

• Keep a consistent top-to-bottom hierarchy: title, KPI summary, main comparison, supporting detail, notes.

• Use the same fuel selector and year conventions across dashboards.

• Limit colour categories and use accessible contrast.

• Use descriptive titles, readable axis units, concise tooltips and source notes.

• Test filter scope: apply a filter to only the intended worksheet when required; use “Apply to Worksheets” settings deliberately.

7.3 Tableau story design

Story point	Narrative	Suggested view
1. Global context	Introduce the 2020–2026 dataset scope and fuel types.	Overview dashboard / KPI tiles
2. Time trends	Show how each fuel type changes across years.	Trend line chart
3. Country leaders	Identify the highest and lowest petrol-price countries for a selected year.	Top/bottom country ranking
4. Regional differences	Compare regional averages and within-region variation.	Regional chart / map
5. Policy context	Explore tax and subsidy measures alongside price levels.	Scatter plot / subsidy comparison
6. Takeaways	Summarize observed patterns, caveats and future analysis needs.	Conclusion text + selected dashboard

8. Functional and Performance Testing

Field	Value
Student / Individual	ABIR MENDHALKAR
Individual ID	IND-ABIR-055(self-assigned working ID; replace if institution issues an official ID)
Project	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Project period	2020–2026
Dataset scope	27,468 records • 84 countries • 7 global regions • 10 variables (as supplied in project brief)
Prepared on	1 November 2026

8.1 Tableau performance test plan

Parameter	Test procedure	Evidence / result
Data rendered	Confirm connected source, number of tables and row count.	Capture data-source page and row count; target per brief is 27,468 rows, not independently verified.
Data preprocessing	Show null, duplicate, standardization and cleaning checks.	Attach query outputs and before/after counts.
Filters	Test Year, Country, Region and Fuel Type parameter independently and in combination.	Capture each selection and verify all intended worksheets update.
Calculated fields	Validate Average Fuel Price and any change/ranking calculations against hand-checked sample rows.	Show formula and sample reconciliation.
Dashboard design	Confirm three dashboards and count visualizations per dashboard.	Attach full-screen screenshots; count actual worksheets.
Story design	Confirm story sequence and all story points.	Attach Tableau story screenshot and public URL if published.

8.2 Snowflake performance test plan

Parameter	Test procedure	Evidence / result
Data rendered	Record database, schema, table, rows and storage size.	Snowflake table details and count query.
Data quality	Run null, duplicate, trim, standardization and cleaning checks.	SQL outputs and query IDs.
New fields	List derived columns and business definitions.	DDL/view definition and sample output.
Filter efficiency	Compare query elapsed time with and without selective filters on the same warehouse.	Query IDs, elapsed time and warehouse size; repeat tests for fairness.
Visualization testing	Record number of visualizations and dashboards connected to curated data.	Tableau screenshots.
Query history	Identify the longest query and inspect its profile.	Query history screenshot and query ID.

No live Snowflake account, query history, workbook or execution screenshots were included. Therefore actual timings, storage sizes, query IDs and dashboard counts cannot be truthfully populated.

9. Code and Solution Summary

Field	Value
Student / Individual	ABIR MENDHALKAR
Individual ID	IND-ABIR-055(self-assigned working ID; replace if institution issues an official ID)
Project	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Project period	2020–2026
Dataset scope	27,468 records • 84 countries • 7 global regions • 10 variables (as supplied in project brief)
Prepared on	1 November 2026

9.1 Project / repository details

Field	Completed entry
Project / repository link	https://github.com/abir22-wq/globalfuelprices
Dataset source link	https://drive.usercontent.google.com/download?
Database	id=13k0qQW8OqvyIrE3yCzAym5SUTmQNd0qu&export=download&authuser=0
Warehouse	Proposed: FUEL_INTELLIGENCE_DB (use actual database name if created).
Tables / views	Not supplied. Add actual Snowflake warehouse name and size if used.
Key transformations	Proposed logical names: RAW_FUEL_PRICES, STG_FUEL_PRICES, CURATED_FUEL_PRICES. Trim/standardize labels; validate years and numeric values; profile nulls/duplicates; derive fuel type and
KPIs	average price measures.
Key insights	Average fuel price by fuel type, country, region and year; top/bottom country rankings; year-over-year change if validated.
	To be populated only after workbook results are generated and checked; do not insert assumed country rankings or trends.

9.2 Reproducibility checklist

Criteria	Status
Source data loaded and accessible	Not verified — source file / environment not supplied
Data cleaned and transformed	Planned; execution evidence required
Tables, views and relationships implemented	Not verified
SQL logic and KPIs implemented	Formulas specified; implementation evidence required
Outputs validated against requirements	Pending sample reconciliation
Solution organized and documented	Documentation prepared in this consolidated pack; project files still required

10. Advantages, Disadvantages and Conclusion

Field	Value
Student / Individual	ABIR MENDHALKAR
Individual ID	IND-ABIR-055 (self-assigned working ID; replace if institution issues an official ID)
Project	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Project period	2020–2026
Dataset scope	27,468 records • 84 countries • 7 global regions • 10 variables (as supplied in project brief)
Prepared on	1 November 2026

10.1 Advantages

• Consolidates multi-country and multi-year comparisons in one interactive interface.

• Allows switching between Petrol, Diesel and LPG without rebuilding separate dashboards.

• Supports country rankings, regional comparisons and trend exploration.

• Makes data limitations and policy context visible alongside charts.

• Can be extended with additional years, countries and official data sources.

10.2 Disadvantages / limitations

• The analysis is only as reliable as the source definitions, coverage and update frequency.

• Cross-country comparisons may be distorted by differences in currency, unit, tax treatment or reporting period if not normalized.

• Descriptive charts do not establish causal effects of taxes or subsidies.

• Live refresh, predictive modelling and public deployment are not confirmed in the supplied materials.

• No actual Tableau screenshots, Snowflake outputs or tested performance metrics were provided.

10.3 Conclusion

The proposed Global Fuel Prices and Taxation Trends BI solution addresses the challenge of exploring fuel-price and policy-context data across countries, regions, years and fuel types. The completed design specifies the customer problem, requirements, architecture, data-preparation workflow, eight business questions, three dashboards, a story sequence and a validation plan. Final empirical conclusions must be added after loading the source dataset, executing quality checks and validating the Tableau outputs.

10.4 Future scope

Phase	Enhancement	Indicative timeline	Expected impact
Phase 2	Automated refresh from trusted official data sources and source metadata tracking.	Next iteration	More current data and reduced manual loading.
Phase 3	Currency/unit normalization and documented purchasing-power or affordability measures.	After data governance review	Fairer comparisons across countries.
Phase 4	Forecasting and scenario analysis with validated historical coverage and uncertainty intervals.	Advanced extension	Exploratory forward-looking analysis; not a substitute for policy evaluation.
Phase 5	Role-based access, scheduled reporting and multilingual dashboard labels.	Deployment phase	Broader adoption and maintainability.

11. Communication and Demonstration Plan

Field	Value
Student / Individual	ABIR MENDHALKAR
Individual ID	IND-ABIR-055 (self-assigned working ID; replace if institution issues an official ID)
Project	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Project period	2020–2026
Dataset scope	27,468 records • 84 countries • 7 global regions • 10 variables (as supplied in project brief)
Prepared on	1 November 2026

11.1 Communication plan

Audience	Message	Channel / format	Frequency
Project evaluator	Scope, methodology, evidence and limitations	Final report and live demo	At submission
Energy-policy / research user	Country, region, year and fuel comparisons	Interactive dashboard and short briefing	When analysis is updated
Project supervisor	Progress, blockers and validation status	Progress note / review meeting	At sprint review
General viewer	How to interpret charts and filters	Dashboard captions, tooltips and source notes	Embedded in dashboard

11.2 Proposed demo plan (individual project)

No.	Demo section	Description	Duration	
1	Introduction	Introduce ABIR MENDHALKAR, project title, dataset scope and objective.	1 min	
2	Problem statement	Explain fragmented comparisons and user needs.	1 min	
3	Data preparation	Explain source scope, quality checks and transformations; show genuine SQL outputs.	2 min	
4	Dashboard overview	Navigate the three dashboards and explain key views.	3 min	
5	Interactive features	Change Fuel Type, Year, Region and Country filters; demonstrate top-three petrol ranking.	2 min	
6	Conclusion and Q&A;	Summarize findings, limitations and future scope.	1 min	

11.3 Demo flow summary

Step	Activity	Notes
1	Introduction & problem statement	State audience, project objective and dataset coverage.
2	Solution overview	Show the logical data flow and dashboard structure.
3	Live feature demonstration	Use the Fuel Type parameter and filters; explain one trend and one ranking.
4	Q&A; session	Answer from displayed evidence; distinguish measured findings from future scope.

11.4 Team involvement in demonstration

Member	Role	Section presented	Contribution	Participation
SIDDHI SAMDANI	Individual project presenter / analyst	All sections	Prepared documentation, explains data workflow, dashboards, testing evidence and limitations.	To be confirmed during actual demo

Team coordination: Individual submission; no team leader or additional members supplied. Demo issues and resolutions should be recorded after the actual presentation, not pre-filled as events that have not occurred.

12. Appendix: Deliverables and Evidence Checklist

Field	Value
Student / Individual	SIDDHI SAMDANI
Individual ID	IND-SIDDHI-001 (self-assigned working ID; replace if institution issues an official ID)
Project	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Project period	2020–2026
Dataset scope	27,468 records • 84 countries • 7 global regions • 10 variables (as supplied in project brief)
Prepared on	1 November 2026

12.1 Files/evidence to attach before final academic submission

Item		What to add if available
Dataset file and source URL		Original dataset and official source/citation.
Tableau workbook (.twb/.twbx)		Working workbook with 8 worksheets, 3 dashboards and story.
Dashboard screenshots		Clear screenshots of each completed dashboard and key interactions.
Snowflake SQL scripts / outputs	Illustrative query patterns provided	Actual scripts, query outputs, query IDs and table/view definitions.
Performance evidence	Test plan included	Measured query times, row counts, query history and Tableau responsiveness.
GitHub / demo links	https://github.com/abir22-wq/globalfuelprices	Real public/private links with appropriate access.
Communication / demo evidence	https://drive.google.com/file/d/1MXxkOpexH0lpbBqi6cKdJ7ePGCvcPMfv/view?usp=drive_link	Presentation slides, actual demo recording/link if required, and attendance/participation confirmation.